

Notes: Glitz & Meyersson (AER, 2020)

Industrial Espionage and Productivity

Contents

1	Big picture	3
2	Data and measurement (key objects)	3
2.1	Intelligence data: “pieces of information” and sector assignment	3
2.2	Industry outcomes: output, inputs, TFP, trade, and R&D proxies	4
3	Models and empirical specifications (all equations + interpretation)	4
3.1	A diffusion-style productivity growth equation (Eqs. (1)–(3))	4
3.2	The main “gap” regression (Eq. (4))	5
3.3	Measuring productivity (Eq. (5))	5
3.4	A “one-step” alternative outcome specification (Eq. (6))	6
4	Identification and instruments (what makes the estimates credible)	6
4.1	Baseline identification idea	6
4.2	IV strategy (Table 3): two instruments for $S_{j,t}^E/Y_{j,t}^E$	6
5	Tables and figures	7
5.1	Figure 1: sectoral distribution of industrial intelligence	7
5.2	Table 1: summary statistics (levels and changes)	8
5.3	Figure 2: two illustrative case studies (micro evidence)	8
5.4	Table 2 and Figure 3: baseline OLS relationship (TFP gap)	8
5.5	Figure 4 and Table 3: IV logic and results	9
5.6	Table 4: decomposition (where does the gap closing come from?)	10
5.7	Table 5: robustness (measurement and omitted-variable checks)	10
5.8	Figure 5: placebo/randomization inference	11
5.9	Table 6: alternative outcomes (beyond TFP)	11
5.10	Table 7: quality of intelligence (fat tails in impact)	12
5.11	Figure 6: heterogeneous effects (when does espionage matter most?)	12
5.12	Figure 7: counterfactual “no espionage” productivity path	13
6	Short summary	13

7	Conclusion	14
7.1	Core conclusion (what the paper establishes)	14
7.2	Economic intuition	14
7.3	Takeaway	15

1 Big picture

- **Question:** Does *state-led industrial espionage* generate economically meaningful technology diffusion?
- **Setting:** East Germany (GDR) ran a large foreign intelligence operation (HVA/Stasi) to acquire proprietary Western (FRG) industrial and technological information.
- **Core empirical object:** a sector-by-time measure of *industrial espionage inflow* (counts of “pieces of information” about West German industry, classified into 16 sectors), scaled by East German sector output.
- **Main result:** higher espionage inflows predict *faster East German productivity growth relative to West Germany* (i.e., a narrowing of the FRG–GDR sectoral TFP gap).
- **Mechanism/intuition:** espionage is a (partly illegal) technology-transfer channel; it is especially powerful when the recipient has sufficient absorptive capacity (closer to frontier) and when legal channels (e.g., imports) are constrained.
- **Why we care:** this is a clean example where a *non-market information flow* moves *TFP gaps* in a way consistent with diffusion models. It also illustrates measurement/identification challenges when “technology” is endogenous and partially hidden.

2 Data and measurement (key objects)

2.1 Intelligence data: “pieces of information” and sector assignment

- Source: the Stasi’s SIRA database (subdatabase 11), which records metadata on industrial intelligence.
- Coverage: 189,725 pieces of information between 1968–1989; 151,627 can be allocated to the 16-sector classification used in the analysis.
- Each item contains:
 - a textual description and a set of keywords (the keyword universe is large: 143,005 distinct keywords),
 - an origin (source/informant),
 - sometimes a quality evaluation by analysts (but often missing).
- Baseline sector assignment: map the 2,000 most common keywords to sectors using the German Standard Industrial Classification; assign an item to a sector if it contains at least one keyword mapped to that sector.
- Alternative assignments (robustness, Table 5):

- *Keyword-weighted inflow*: weight each intelligence item by the number of sector-mapped keywords it contains.
- *Machine-learning inflow*: train a support vector machine (SVM) text classifier using manually categorized items; use predicted sector probabilities to allocate items.

2.2 Industry outcomes: output, inputs, TFP, trade, and R&D proxies

- Main sector outcomes come from Heske’s harmonized FRG/GDR industry accounts.
- Additional outcomes come from East German Statistical Yearbooks:
 - physical quantities of 104 products (15 of 16 sectors),
 - product variety measures and export measures used in Table 6.
- R&D proxy R_{ijt} : sectoral patent applications (GDR and FRG), scaled by sector output.
- Trade proxy: import intensity (imports/output), and the FRG–GDR import-intensity gap.

3 Models and empirical specifications (all equations + interpretation)

3.1 A diffusion-style productivity growth equation (Eqs. (1)–(3))

Let A_{ijt} denote TFP in sector i and country j at time t . The paper motivates a reduced-form productivity-growth equation where productivity growth depends on changes in “knowledge stocks” and distance to a technological frontier A_{jt}^F :

$$\Delta \ln A_{ijt+1} = \beta \Delta \ln E_{ijt} + \eta \Delta \ln G_{ijt} + \theta \ln \left(\frac{A_{jt}^F}{A_{ijt}} \right) + \xi_{ijt+1}, \quad (1)$$

where (conceptually)

- E_{ijt} is an *espionage-related* knowledge stock,
- G_{ijt} is an *R&D-related* knowledge stock,
- $\ln(A_{jt}^F/A_{ijt})$ is the (log) distance to the frontier (a convergence force).

A convenient reparameterization expresses marginal effects in “per output” units:

$$\Delta \ln A_{ijt+1} = \rho \frac{\Delta E_{ijt}}{Y_{ijt}} + \eta \frac{\Delta G_{ijt}}{Y_{ijt}} + \theta \ln \left(\frac{A_{jt}^F}{A_{ijt}} \right) + \xi_{ijt+1}, \quad (2)$$

where Y_{ijt} is sector output and ρ can be interpreted as the *marginal productivity of espionage knowledge*.

Empirical counterparts. The paper maps these unobserved changes into observables:

- ΔE_{ijt} is proxied by the *inflow of intelligence* S_{ijt} (counts of new “pieces of information”),
- ΔG_{ijt} is proxied by sectoral *patent applications* R_{ijt} .

This yields the core estimating equation:

$$\Delta \ln A_{ijt+1} = \rho \frac{S_{ijt}}{Y_{ijt}} + \eta \frac{R_{ijt}}{Y_{ijt}} + \theta \ln \left(\frac{A_{ijt}^F}{A_{ijt}} \right) + \lambda_{ij} + \pi_{it+1} + \mu_{jt+1} + \varepsilon_{ijt+1}. \quad (3)$$

Fixed effects intuition (Eq. (3)):

- λ_{ij} : sector–country fixed effects (permanent level differences),
- π_{it+1} : sector–time effects (global sector shocks common across countries),
- μ_{jt+1} : country–time effects (country-wide shocks common across sectors).

3.2 The main “gap” regression (Eq. (4))

For identification, the paper focuses on the *West–East productivity gap*. Assuming West Germany is the sectoral frontier, Eq. (3) can be transformed into:

$$\Delta \ln \left(\frac{A_{j,t+1}^W}{A_{j,t+1}^E} \right) = -\rho \frac{S_{j,t}^E}{Y_{j,t}^E} + \eta \left(\frac{R_{j,t}^W}{Y_{j,t}^W} - \frac{R_{j,t}^E}{Y_{j,t}^E} \right) - \theta \ln \left(\frac{A_{j,t}^W}{A_{j,t}^E} \right) + \lambda_j + \pi_{t+1} + \varepsilon_{j,t+1}. \quad (4)$$

Key sign prediction: if espionage raises East German productivity, then $S_{j,t}^E$ should *reduce* the growth of the gap; thus the coefficient on $S_{j,t}^E/Y_{j,t}^E$ should be **negative** (because of the leading minus sign).

3.3 Measuring productivity (Eq. (5))

TFP is constructed from a Cobb–Douglas production function:

$$\ln A_{ijt} = \ln Y_{ijt} - \alpha_{ij} \ln K_{ijt} - \beta_{ij} \ln L_{ijt}, \quad \beta_{ij} = 1 - \alpha_{ij}. \quad (5)$$

Implementation notes:

- Y_{ijt} is gross value added (real), K_{ijt} is the real capital stock, and L_{ijt} is employment.
- Capital stocks are constructed using a perpetual inventory method.
- Factor shares: FRG shares come from national accounts; GDR shares are proxied using capital shares from a set of post-socialist EU economies (details in the paper).

3.4 A “one-step” alternative outcome specification (Eq. (6))

To check that results are not an artifact of the two-step TFP construction, the paper also estimates an output-growth version:

$$\Delta \ln \left(\frac{Y_{j,t+1}^W}{Y_{j,t+1}^E} \right) = -\rho \frac{S_{j,t}^E}{Y_{j,t}^E} + \eta \left(\frac{R_{j,t}^W}{Y_{j,t}^W} - \frac{R_{j,t}^E}{Y_{j,t}^E} \right) - \theta \ln \left(\frac{A_{j,t}^W}{A_{j,t}^E} \right) + \alpha_j \Delta \ln \left(\frac{K_{j,t+1}^W}{K_{j,t+1}^E} \right) + \beta_j \Delta \ln \left(\frac{L_{j,t+1}^W}{L_{j,t+1}^E} \right) + \varepsilon_{j,t+1}. \quad (6)$$

Interpretation: Eq. (6) directly relates relative output growth to espionage inflows, controlling for relative input growth. It is used when outcomes are not TFP (e.g., quantity output, product variety, exports in Table 6).

4 Identification and instruments (what makes the estimates credible)

4.1 Baseline identification idea

Eq. (4) is a sector-by-time panel regression where the identifying variation is:

- *within-sector over time* changes in espionage inflow intensity, after controlling for sector FE and time FE,
- interpreted as plausibly exogenous “technology shocks” originating from the intelligence apparatus.

The key concern is **endogeneity**: the GDR might target espionage toward sectors that are already poised to grow (or toward sectors where it is falling behind), which would bias OLS.

4.2 IV strategy (Table 3): two instruments for $S_{j,t}^E/Y_{j,t}^E$

Instrument 1: “Old informants” (shift-share/Bartik logic). Let I_s be the total inflow in period s received from informants already active in 1970. Define weights:

- $\theta_{i,70}$: informant i ’s share of total inflow in 1970,
- $\lambda_{ij,70}$: share of informant i ’s 1970 inflow allocated to sector j .

Then the predicted sector inflow intensity is:

$$Z_{j,t}^{old} \equiv \frac{\sum_{i \in 1970} \theta_{i,70} \lambda_{ij,70} \left(\sum_{s=t-2}^t I_s \right)}{Y_{j,t}^E}. \quad (1)$$

Intuition: sector exposure is “frozen” at its 1970 composition; time variation comes from aggregate changes in productivity/activity of informants already operating in 1970.

Instrument 2: “Exit of informants”. Let $i^*(s)$ be the set of informants last observed in period s (they “exit”). Let $\bar{I}_{i^*j}$ be informant i^* ’s average annual inflow related to sector j over the full sample. The exit-based predicted loss in inflow intensity is constructed as:

$$Z_{j,t}^{exit} \equiv \frac{\sum_{s=t-5}^{t-3} \sum_{i^*(s): \bar{I}_{i^*j} \geq 20} \bar{I}_{i^*j}}{Y_{j,t}^E}. \quad (2)$$

Intuition: when prolific informants disappear, sector-specific espionage inflows fall mechanically; if exits are not driven by contemporaneous sector productivity shocks, this isolates plausibly exogenous variation.

Inference note. The paper clusters standard errors by sector (16 clusters) and reports wild cluster bootstrap- t p -values to handle few clusters.

5 Tables and figures

5.1 Figure 1: sectoral distribution of industrial intelligence

Read:

- Espionage is *not uniform* across the economy: it concentrates heavily in electronics / office equipment / computers.
- This concentration is helpful empirically: there is large cross-sector variation in the key regressor.
- Economically, it suggests a revealed preference: the intelligence apparatus prioritized high-tech sectors with potentially large productivity payoffs.

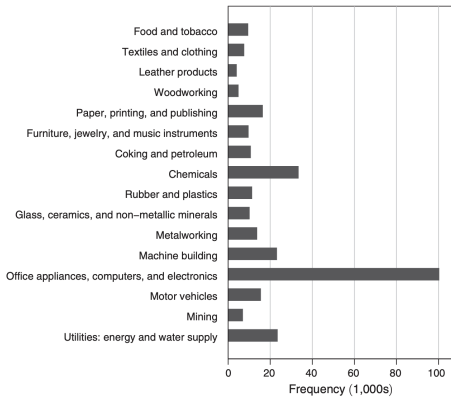


FIGURE 1. SECTORAL DISTRIBUTION OF INFORMATION

Note: Figure shows the sector-specific inflows of information received by the HVA between 1968 and 1989.

	West Germany		East Germany		Difference	
	Mean (1)	SD (2)	Mean (3)	SD (4)	Mean (5)	SD (6)
Inflow/Y			1.524	(1.403)		
Capital share	0.282	(0.152)	0.399	(0.146)		
$\Delta \log$ TFP	0.028	(0.070)	0.038	(0.069)	−0.010	(0.097)
$\Delta \log$ output per worker	0.050	(0.079)	0.090	(0.070)	−0.040	(0.098)
Patents/Y	0.392	(0.372)	0.313	(0.433)	0.079	(0.225)
$\log$ TFP	2.349	(0.636)	0.859	(0.605)	1.490	(0.923)
$\log$ output per worker	3.679	(0.414)	2.385	(1.091)	1.294	(0.782)
Imports/Y	2.562	(4.609)	0.312	(0.333)	2.250	(4.622)

Notes: Unweighted summary statistics computed for 3-year overlapping observations for the period 1970 to 1989 apart from the capital shares for East Germany which are based on the period 1995 to 2006 and refer to the aggregate capital shares in the ten new EU member states who joined in May 2004. Imports are cumulated over the last 3 years and measured in million dollars at constant 1995 prices. Output is measured in million euros at constant 1995 prices. Workers are measured in 1,000 so that output per worker is measured in 1,000 euros at constant 1995 prices. The number of observations 240 (234 for Imports/Y, 320 for the West German capital share, and 192 for the East German capital share).

5.2 Table 1: summary statistics (levels and changes)

Read:

- The **levels** show a large initial gap: log TFP and log output per worker are much higher in FRG than GDR.
- Over the sample, the **changes** suggest catch-up: mean $\Delta \log$ TFP (and especially $\Delta \log$ output per worker) is higher in GDR.
- Import intensity is dramatically higher in FRG; the GDR is much more closed (relevant for the “substitution for trade” interpretation).
- Espionage inflow intensity has substantial dispersion (mean ≈ 1.52 , s.d. ≈ 1.40), so regression coefficients can map into meaningful effect sizes.

5.3 Figure 2: two illustrative case studies (micro evidence)

Read:

- The figure visualizes that intelligence inflows are “bursty” in particular technologies.
- In both examples, bursts of intelligence appear around periods when GDR production ramps up, consistent with the mechanism that information enables adoption/production.
- This is *suggestive* (not causal by itself), but it makes the macro regression story more believable.

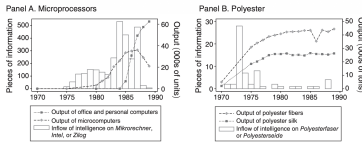


FIGURE 2. CASE STUDIES

Source: SIRA subdatabase 11 for the number of pieces of information that include either of the keywords *Intel*, *Zilog*, or *Mikroware* (microcomputer) in panel A or *Polyesterfaser* (polyester fibers) or *Polyesterseide* (polyester silk) in panel B. For output, figures are taken from the statistical yearbooks of the German Democratic Republic. Output figures refer to the industrial production of office and personal computers (*Büro- und Personalcomputer*) and microcomputers (*Mikrowarechner*) in panel A and polyester fibers (*Polyesterfaser*) and polyester silk (*Polyesterseide*) in panel B.

	log TFP			log output per worker		
	Baseline spec (1)	Patents gap (2)	Lagged gap (3)	Baseline spec (4)	Patents gap (5)	Lagged gap (6)
Espionage	-0.034 (0.021)	-0.041 (0.021)	-0.052 (0.012)	-0.030 (0.018)	-0.040 (0.017)	-0.039 (0.017)
Patents gap		0.071 (0.028)	-0.038 (0.034)		0.103 (0.026)	0.012 (0.028)
log TFP gap			-0.564 (0.090)			
log output/worker gap					-0.514 (0.100)	
p-value WB	0.199	0.128	0.011	0.164	0.072	0.125
R ²	0.34	0.36	0.56	0.31	0.35	0.51
Observations	240	240	240	240	240	240

Notes: Sample based on 3-year intervals and overlapping observations for the period 1970 to 1989. All regressions include time- and sector-specific fixed effects. Observations are weighted by the average number of workers in a sector. The dependent variable is the change in the log TFP gap between West and East Germany over the period t to $t+3$ in columns 1 to 3 and the change in the log output per worker gap over the period t to $t+3$ in columns 4 to 6. Standard errors are clustered at the sectoral level and shown in parentheses. p-value WB denotes p-values, relating to the espionage estimate, from Cameron, Gelbach, and Miller's (2008) wild cluster bootstrap procedure using 999 replications.

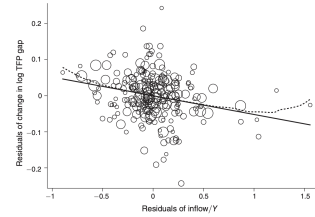


FIGURE 3. INDUSTRIAL ESPIONAGE AND PRODUCTIVITY

Notes: The figure plots residualized changes in the log TFP gap between West and East Germany against residualized sector-specific inflows of information on the basis of the specification reported in column 3 of Table 2. Circles are proportional to the square root of the average number of workers in an industry. The solid black line represents the OLS regression line and the dashed line the fit from a local polynomial regression.

5.4 Table 2 and Figure 3: baseline OLS relationship (TFP gap)

Read:

- In Table 2, the coefficient on $\frac{S_{j,t}^E}{Y_{j,t}^E}$ is **negative and statistically significant** for both:
 - the **TFP gap** $\Delta \log(A^W/A^E)$ (columns 1–3), and
 - the **output-per-worker gap** $\Delta \log((Y/L)^W/(Y/L)^E)$ (columns 4–6).
- Economically: using the most saturated TFP-gap specification (col. 3), the point estimate is about -0.052 .

- A 1-unit increase in espionage intensity reduces the *3-year change* in the log TFP gap by about 0.052 (roughly a 5% narrowing of the ratio).
- A 1 s.d. increase (≈ 1.40) implies a gap change of about 0.073 (about 7%).
- The lagged gap enters with a large negative coefficient: strong **conditional convergence** (catch-up when initially behind).
- Estimation details that matter for interpretation:
 - 3-year intervals with *overlapping* observations (rolling windows),
 - employment weights,
 - inference with sector clustering (16 clusters) and wild cluster bootstrap-*t* *p*-values.
- Figure 3 shows the same relationship in residualized form: after partialling out fixed effects/controls, higher inflows line up with larger gap reductions.

5.5 Figure 4 and Table 3: IV logic and results

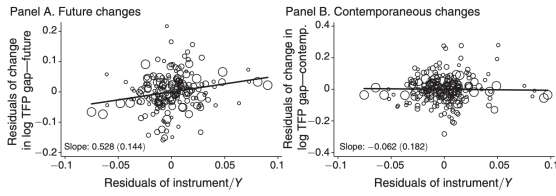


FIGURE 4. EXITS OF INFORMANTS AND CHANGES IN THE LOG TFP GAP

Notes: The figure plots residualized changes in the log TFP gap between West and East Germany against residualized exits of highly prolific informants scaled by output. Exits are measured between the end of period $t-6$ and $t-3$. Changes in the log TFP gap are measured between the end of period t and $t+3$ in panel A, and the end of $t-6$ and $t-3$ in panel B. Circles are proportional to the square root of the average number of workers in an industry. The solid black lines represent the OLS regression lines.

TABLE 3—INSTRUMENTAL VARIABLES

	log TFP				log output per worker			
	Old informants		Exit of informants		Old informants		Exit of informants	
	First stage (1)	IV results (2)	First stage (3)	IV results (4)	First stage (5)	IV results (6)	First stage (7)	IV results (8)
Espionage		-0.072 (0.024)		-0.120 (0.036)		-0.059 (0.028)		-0.119 (0.040)
Patents gap	-0.569 (0.392)	-0.034 (0.024)	-0.014 (0.232)	-0.059 (0.048)	-0.528 (0.387)	0.018 (0.025)	0.017 (0.196)	0.001 (0.049)
log TFP gap	0.009 (0.446)	-0.571 (0.094)	0.544 (0.566)	-0.679 (0.140)				
log output/ worker gap					0.299 (0.305)	-0.514 (0.096)	0.844 (0.383)	-0.613 (0.133)
Instrument old informants	0.631 (0.081)				0.639 (0.082)			
Instrument exits			-4.395 (0.619)				-4.568 (0.536)	
<i>p</i> -value WB		0.150		0.149		0.353		0.188
<i>F</i> -statistic	61.4		50.4		60.5		72.6	
Observations	240	240	192	192	240	240	192	192

Notes: Sample based on 3-year intervals and overlapping observations for the period 1970 to 1989. All regressions include time- and sector-specific fixed effects. Observations are weighted by the average number of workers in a sector. The dependent variable is the change in the log TFP gap between West and East Germany over the period t to $t+3$ in columns 1 to 4 and the change in the log output per worker gap over the period t to $t+3$ in columns 5 to 8. In columns 1, 2, 5, and 6, the instrument is constructed as $(\sum_{i=1970}^{1989} \theta_{i,70} \lambda_{i,70} \sum_{s=t-6}^{t-3} I_{ij})/Y_{it}^w$, where $\theta_{i,70}$ is the share of the total information received in 1970 sent by informant i , $\lambda_{i,70}$ is the fraction of that information pertaining to sector j , and I_{ij} is the total inflow in period s received from sources already active in 1970. In columns 3, 4, 7, and 8, the instrument is constructed as $(\sum_{s=t-6}^{t-3} s \sum_{i=1970}^{1989} \theta_{i,s} I_{ij})/Y_{it}^w$, where I_{ij} is the average annual inflow of information generated by informant i pertaining to sector j over the entire sample period, and i 's denotes all informants who are last observed in period s . Standard errors are clustered at the sectoral level and shown in parentheses. *p*-value WB denotes *p*-values, relating to the espionage estimate, from Cameron, Gelbach, and Miller's (2008) wild cluster bootstrap-*t* procedure using 999 replications.

Read:

- Figure 4 supports instrument relevance/exogeneity in reduced form:
 - Informant exits predict *future* declines in sector inflows (strong first-stage intuition).
 - Exits do *not* show a strong contemporaneous relationship with TFP-gap changes (comfort against reverse causality).
- Table 3 first stages are strong (high *F*-stats), and the IV point estimates are *more negative* than OLS:
 - Interpretable as either attenuation bias in OLS (measurement error in S/Y) or as IV isolating “cleaner” variation.

5.6 Table 4: decomposition (where does the gap closing come from?)

Read:

- The gap is $\Delta \ln(A^W/A^E) = \Delta \ln A^W - \Delta \ln A^E$.
- Table 4 shows the espionage effect is driven mainly by **higher GDR TFP growth** (positive effect on $\Delta \ln A^E$), with little systematic effect on FRG.
- This supports the key mechanism: espionage operates as technology adoption in the recipient (GDR), rather than depressing the frontier (FRG).

TABLE 4—DECOMPOSITIONS

	Baseline decomposition			Flexible decomposition		
	FRG/GDR (1)	FRG (2)	GDR (3)	FRG/GDR (4)	FRG (5)	GDR (6)
<i>OLS</i>						
Espionage	−0.052 (0.012)	−0.009 (0.005)	0.043 (0.012)	−0.036 (0.009)	−0.010 (0.005)	0.046 (0.012)
Patents gap	−0.038 (0.024)	0.004 (0.019)	0.043 (0.020)			
log TFP gap	−0.564 (0.090)	−0.208 (0.069)	0.356 (0.058)	−0.594 (0.088)		0.319 (0.051)
GDR patents/Y				−0.083 (0.057)		−0.022 (0.038)
FRG patents/Y				−0.147 (0.069)	0.073 (0.028)	
<i>p</i> -value WB	0.011	0.109	0.011	0.000	0.035	0.001
<i>R</i> ²	0.56	0.66	0.45	0.57	0.61	0.44
Observations	240	240	240	240	240	240
<i>IV: Old informants</i>						
Espionage	−0.072 (0.024)	−0.025 (0.018)	0.047 (0.014)	−0.023 (0.038)	−0.031 (0.017)	0.068 (0.016)
Patents gap	−0.034 (0.024)	0.007 (0.020)	0.042 (0.018)			
log TFP gap	−0.571 (0.094)	−0.214 (0.067)	0.357 (0.055)	−0.594 (0.074)		0.335 (0.057)
GDR patents/Y				−0.099 (0.079)		−0.032 (0.047)
FRG patents/Y				−0.164 (0.083)	0.091 (0.039)	
<i>p</i> -value WB	0.150	0.332	0.030	0.565	0.135	0.036
<i>F</i> -statistic	61.7	61.7	61.7	14.1	14.1	52.5
Observations	240	240	240	240	240	240
<i>IV: Exit of informants</i>						
Espionage	−0.120 (0.036)	−0.035 (0.017)	0.085 (0.028)	−0.090 (0.038)	−0.024 (0.017)	0.090 (0.034)
Patents gap	−0.059 (0.048)	−0.005 (0.029)	0.054 (0.034)			
log TFP gap	−0.679 (0.140)	−0.278 (0.102)	0.402 (0.090)	−0.706 (0.127)		0.368 (0.081)
GDR patents/Y				−0.029 (0.088)		−0.059 (0.064)
FRG patents/Y				−0.156 (0.118)	0.094 (0.050)	
<i>p</i> -value WB	0.149	0.143	0.320	0.109	0.236	0.362
<i>F</i> -statistic	50.4	50.4	50.4	35.3	70.8	31.2
Observations	192	192	192	192	192	192

Notes: Sample based on 3-year intervals and overlapping observations for the period 1970 to 1989. All regressions include time- and sector-specific fixed effects. Observations are weighted by the average number of workers in a sector. The dependent variable is the change in the log TFP gap between West and East Germany over the period t and $t + 3$ in columns 1 and 4 and the change in log TFP between t and $t + 3$ in West and East Germany in columns 2 and 5 and 3 and 6, respectively. The instrumental variables are described in Section VA. Standard errors are clustered at the sectoral level and shown in parentheses. *p*-value WB denotes *p*-values, relating to the espionage estimate, from Cameron, Gelbach, and Miller's (2008) wild cluster bootstrap-*t* procedure using 999 replications.

TABLE 5—ROBUSTNESS: LOG TFP

	Main spec (1)	Weighted by output (2)	No weights (3)	No IT (4)	Sector trends (5)	Trade gap (6)	Keyword weighted (7)	Machine learning (8)
<i>OLS</i>								
Espionage	−0.052 (0.012)	−0.077 (0.032)	−0.047 (0.013)	−0.047 (0.016)	−0.044 (0.012)	−0.053 (0.012)	−0.090 (0.024)	−0.049 (0.017)
<i>p</i> -value WB	0.011	0.080	0.028	0.027	0.034	0.008	0.056	0.436
<i>R</i> ²	0.56	0.59	0.54	0.54	0.72	0.57	0.56	0.55
Observations	240	240	240	225	240	234	240	240
<i>IV: Old informants</i>								
Espionage	−0.072 (0.024)	−0.116 (0.048)	−0.070 (0.028)	−0.050 (0.042)	−0.009 (0.039)	−0.077 (0.027)	−0.112 (0.039)	−0.081 (0.030)
<i>p</i> -value WB	0.150	0.188	0.070	0.296	0.841	0.174	0.192	0.241
<i>F</i> -statistic	61.4	59.5	23.9	37.8	16.8	57.1	24.2	18.6
Observations	240	240	240	225	240	234	240	240
<i>IV: Exit of informants</i>								
Espionage	−0.120 (0.036)	−0.168 (0.072)	−0.109 (0.040)	−0.144 (0.078)	−0.213 (0.068)	−0.117 (0.035)	−0.275 (0.086)	−0.118 (0.031)
<i>p</i> -value WB	0.149	0.241	0.041	0.036	0.006	0.163	0.150	0.087
<i>F</i> -statistic	50.4	53.7	23.3	3.8	11.5	47.1	27.5	27.9
Observations	192	192	192	180	192	189	192	192

Notes: Sample based on 3-year intervals and overlapping observations for the period 1970 to 1989. All regressions include time- and sector-specific fixed effects, the patents gap, and the initial log TFP gap as additional regressors. Observations are weighted by the average number of workers in a sector (apart from columns 2 and 3). The dependent variable is the change in the log TFP gap between West and East Germany over the period t to $t + 3$. Column 1 restates our main results from column 3 of Table 2. In column 2, observations are weighted by the average sector-specific gross value added. In column 3, observations are unweighted. In column 4, we exclude the IT sector from the estimation sample. In column 5, we include sector-specific linear time trends in the specification. In column 6, we include the gap in the sector-specific import/output ratio between West and East Germany as an additional control variable. In column 7, we weight each piece of information according to the number of categorized keywords assigned to each sector. In column 8, we use machine learning methods to assign pieces of information to industry sectors. Standard errors are clustered at the sectoral level and shown in parentheses. *p*-value WB denotes *p*-values, relating to the espionage estimate, from Cameron, Gelbach, and Miller's (2008) wild cluster bootstrap-*t* procedure using 999 replications.

5.7 Table 5: robustness (measurement and omitted-variable checks)

Read:

- Across many redefinitions of the espionage measure (keyword-weighting, SVM mapping) and sample choices (dropping IT, adding sector trends, controlling for import-gap dynamics), the coefficient remains **negative**.
- This reduces the concern that results are driven by:
 - a small set of sectors (e.g., IT),

- slow-moving sector trends correlated with espionage targeting,
 - a particular keyword-mapping rule.
- IV estimates using informant exits are often large in magnitude and significant, consistent with substantial attenuation/measurement error in OLS.

5.8 Figure 5: placebo/randomization inference

Read:

- The placebo coefficients (under random reassignment of intelligence flows) are centered near zero.
- The actual estimate lies in the tail, suggesting the main finding is not a mechanical artifact of the panel structure.

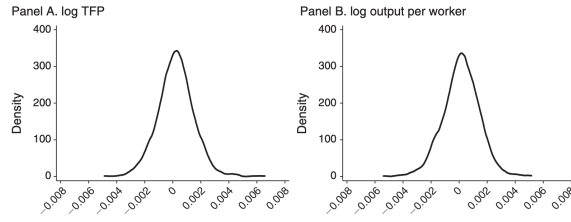


FIGURE 5. PLACEBO TESTS

Notes: The figure shows the distribution of 1,000 estimated coefficients representing the impact of hypothetical espionage inflow rates on changes in the log TFP gap (panel A) and log output per worker gap (panel B) between West and East Germany. Each estimate is obtained by (i) randomly reshuffling the 240 actual espionage inflows S_{it} across observations defined by sector and year, (ii) computing hypothetical espionage inflow rates by dividing the randomly assigned inflows by the actual sector-specific outputs Y_{it} , and (iii) estimating the model by OLS using the same set of control variables as in the main specifications reported in columns 3 and 6 of Table 2. The plotted densities are based on an Epanechnikov kernel function with default bandwidth 0.0003.

TABLE 6—ALTERNATIVE OUTCOMES

	Quantity Output			New Products		Exports		
	(1)	(2)	(3)	Δ (4)	$\Delta \ln$ (5)	World (6)	NATO (7)	Rest world (8)
<i>OLS</i>								
Espionage	0.035 (0.016)	0.036 (0.020)	0.036 (0.020)	1.784 (0.640)	0.033 (0.018)	0.106 (0.063)	0.005 (0.079)	0.148 (0.092)
GDR patents/Y		yes	yes	yes	yes	yes	yes	yes
log TFP gap		yes	yes	yes	yes	yes	yes	yes
p-value WB	0.038	0.097	0.091	0.032	0.078	0.197	0.954	0.243
R ²	0.07	0.07	0.07	0.57	0.55	0.06	0.18	0.04
Observations	1,558	1,558	1,558	240	240	5,757	1,863	3,894
<i>IV—old informants</i>								
Espionage	0.071 (0.028)	0.080 (0.032)	0.081 (0.033)	2.360 (1.487)	0.051 (0.020)	0.164 (0.225)	0.106 (0.349)	0.160 (0.249)
GDR patents/Y		yes	yes	yes	yes	yes	yes	yes
log TFP gap		yes	yes	yes	yes	yes	yes	yes
p-value WB	0.021	0.034	0.030	0.308	0.084	0.640	0.889	0.684
F-statistic	11.1	23.8	31.2	73.5	73.5	95.7	89.3	92.5
Observations	1,558	1,558	1,558	240	240	5,757	1,863	3,894
<i>IV—exit of informants</i>								
Espionage	-0.032 (0.025)	-0.032 (0.024)	-0.027 (0.019)	-1.704 (2.741)	-0.003 (0.038)	1.235 (0.405)	2.364 (0.806)	0.619 (0.336)
GDR patents/Y		yes	yes	yes	yes	yes	yes	yes
log TFP gap		yes	yes	yes	yes	yes	yes	yes
p-value WB	0.257	0.200	0.175	0.630	0.947	0.117	0.085	0.201
F-statistic	10.3	22.5	30.0	32.4	32.4	20.3	15.5	21.6
Observations	1,246	1,246	1,246	192	192	3,834	1,329	2,505

Notes: Sample based on 3-year intervals and overlapping observations for the period 1970 to 1989. All regressions control for changes in the sector-specific log capital stock and employment as in equation (6), and include time- and sector-specific fixed effects. All dependent variables are measured as changes between periods t and $t+3$. The instrumental variables are described in Section VA. In columns 1 to 3, the estimation is based on a balanced sample of 104 products observed consistently in the East German Statistical Yearbooks over the period 1970 to 1989. In each of the three columns, the dependent variable is the product-specific change in the log physical quantity produced. In column 4, the dependent variable is the sector-specific absolute change in the number of distinct products listed in the East German Statistical Yearbooks. In column 5, the dependent variable is the sector-specific change in log exports of East Germany (measured in constant 1995 dollars) using information from all available importing countries included in the World Trade Flows 1962–2000. In columns 7 and 8, the regressions are run separately for importing NATO countries and all other countries respectively. In columns 4 to 8, observations are weighted by the average number of workers in a sector. Standard errors are clustered at the sectoral level and shown in parentheses. p-value WB denotes p-values, relating to the espionage estimate, from Cameron, Gelbach, and Miller's (2008) wild cluster bootstrap-t procedure using 999 replications.

5.9 Table 6: alternative outcomes (beyond TFP)

Read:

- Table 6 swaps the dependent variable away from TFP and toward **real activity** proxies from East German Statistical Yearbooks (physical quantities, product introductions, exports).
- The OLS and “old informants” IV results point to:
 - higher growth in physical production quantities, and
 - more product introductions (in levels and logs).

- Export results are **positive but imprecise** overall (some coefficients are large, but standard errors are also large and significance varies by destination group and instrument).
- Net: these outcomes make it harder to argue the TFP findings are purely a constructed-measure artifact; espionage seems connected to observable production and product mix.

5.10 Table 7: quality of intelligence (fat tails in impact)

Read:

- Most items have missing quality ratings, but among rated items, the distribution is heavily skewed.
- The results indicate that a **small number of very high-quality items drive a disproportionate share of the productivity effect**.
- This fits a “fat-tailed innovation” view: a few blueprints/process documents can be much more valuable than many low-signal reports.

TABLE 7—QUALITY OF INFORMATION

	$\Delta \log \text{TFP gap}$			$\Delta \log \text{output per worker gap}$		
	Main spec (1)	Observed quality (2)	Imputed quality (3)	Main spec (4)	Observed quality (5)	Imputed quality (6)
Espionage	-0.052 (0.012)			-0.039 (0.017)		
Quality: no value		-2.003 (1.728)	-0.030 (0.205)		-1.156 (1.895)	0.100 (0.197)
Quality: low value		-0.034 (0.614)	-0.223 (0.063)		-0.219 (0.660)	-0.258 (0.078)
Quality: average value		0.015 (0.109)	-0.020 (0.036)		-0.030 (0.131)	-0.010 (0.044)
Quality: valuable		0.059 (0.281)	0.136 (0.083)		0.248 (0.390)	0.212 (0.138)
Quality: very valuable		-1.391 (0.692)	-1.570 (0.554)		-1.603 (0.975)	-1.807 (0.621)
Quality: missing		-0.072 (0.021)	-0.060 (0.046)		-0.051 (0.025)	-0.030 (0.047)
p -value WB	0.011			0.125		
p -value WB: no value		0.300	0.871		0.586	0.649
p -value WB: low value		0.952	0.041		0.733	0.048
p -value WB: average value		0.894	0.606		0.865	0.842
p -value WB: valuable		0.839	0.145		0.630	0.291
p -value WB: very valuable		0.161	0.040		0.319	0.059
p -value WB: missing		0.042	0.273		0.161	0.586
R^2	0.56	0.56	0.57	0.51	0.52	0.54
Observations	240	240	240	240	240	240

Notes: Sample based on 3-year intervals and overlapping observations for the period 1970 to 1989. All regressions include time- and sector-specific fixed effects, the patents gap, and the initial log TFP or output per worker gap as additional regressors. Observations are weighted by the average number of workers in a sector. The dependent variable is the change in the log TFP gap between West and East Germany over the period t to $t+3$ in columns 1 to 3 and the change in the log output per worker gap over the period t to $t+3$ in columns 4 to 6. Prior to the imputation procedure, 0.6 percent of the pieces of information in the sample were given a quality assessment of *no value*, 2.3 percent of *low value*, 26.5 percent of *average value*, 9.6 percent of *valuable*, and 1.1 percent of *very valuable*, with the remaining 59.9 percent *missing*. Standard errors are clustered at the sectoral level and shown in parentheses. p -value WB denotes p -values, relating to the espionage estimate, from Cameron, Gelbach, and Miller's (2008) wild cluster bootstrap- t procedure using 999 replications.

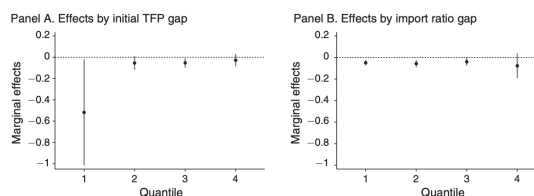


FIGURE 6. HETEROGENEOUS EFFECTS OF INDUSTRIAL ESPIONAGE

Notes: The graphs plot the marginal effects from an OLS specification in which the inflow of information variable is interacted with the quantiles of the initial log TFP gap (panel A) and the initial import intensity gap (panel B). 95 percent confidence intervals are constructed from standard errors clustered at the sectoral level. The corresponding wild cluster bootstrap p -values are 0.111, 0.032, 0.069, and 0.398 for the estimates shown in panel A, and 0.016, 0.014, 0.068, and 0.298 for the estimates shown in panel B.

5.11 Figure 6: heterogeneous effects (when does espionage matter most?)

Read:

- The effect is strongest in sectors with a **smaller initial TFP gap** (i.e., GDR closer to the frontier). I interpret this as *absorptive capacity*: when capabilities are closer to frontier, stolen information is more implementable.
- The effect is also stronger when the **initial import-intensity gap** is larger (GDR more closed relative to FRG), consistent with espionage *substituting for legal trade-based technology transfer*.

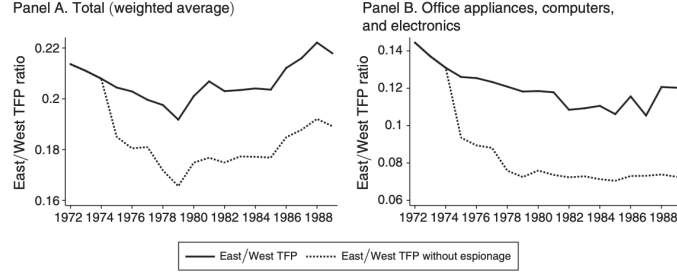


FIGURE 7. COUNTERFACTUAL SIMULATIONS

Notes: The panels depict the counterfactual East/West TFP ratios aggregated across all industry sectors (panel A) and specifically for the *Office Appliances, Computers, and Electronics* sector (panel B). To aggregate across sectors, we take the employment-weighted averages of the actual and counterfactual log TFP gaps before transforming them into East/West TFP ratios. The counterfactual simulations are based on the empirical results reported in column 3 of Table 2.

5.12 Figure 7: counterfactual “no espionage” productivity path

Read:

- The paper uses the estimated model to simulate a counterfactual path setting $S_{j,t}^E = 0$ (no espionage), iterating the implied dynamics forward.
- Aggregate effect (as reported in the text): the observed GDR/FRG TFP ratio rises to about 21.8% by 1989; without espionage it would be about 18.9%.
 - This corresponds to a **13.3% lower** TFP ratio in 1989 in the “no-espionage” counterfactual.
- Sector detail (electronics): the counterfactual gap is especially large in the sector with the largest intelligence inflows, consistent with Figure 1.

6 Short summary

- **Conclusion:** Industrial espionage appears to have been an economically meaningful technology-diffusion channel: sectors that received more Western industrial intelligence experienced faster productivity catch-up in East Germany.
- **Intuition:** espionage increases the recipient’s effective knowledge stock; the gains are largest where the recipient can absorb and implement complex information (nearer to frontier) and where alternative diffusion channels (imports) are restricted.
- **Empirical lesson:** combining unusual micro administrative/intelligence data with a diffusion-style growth regression and careful IV/placebo designs can identify technology transfer effects even in hard-to-measure environments.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Sector-level inflows of foreign industrial intelligence into East Germany (GDR) are associated with *faster productivity catch-up* relative to West Germany (FRG). In the paper’s baseline gap specification, espionage inflows significantly *reduce the growth of the FRG–GDR TFP gap*.
- **Magnitude.** In the main OLS specification for the three-year change in the log TFP gap, the coefficient on espionage intensity is about -0.052 (s.e. ≈ 0.012). A one-standard-deviation increase in the espionage inflow measure (about 1.40) implies an economically meaningful reduction in the growth of the relative TFP gap over the corresponding horizon.
- **Credibility: why this is not just targeting/endogeneity.** The paper argues that endogeneity (the intelligence apparatus targeting certain sectors) is a first-order concern, and it therefore complements OLS with two instrument strategies based on:
 1. a shift-share construction that freezes sector exposures using the 1970 informant-sector composition, and
 2. quasi-exogenous shocks to information supply coming from the exit/disappearance of informants.

Both IV approaches yield estimates that are *larger in magnitude* than OLS, consistent with attenuation from measurement error or with IV isolating cleaner exogenous variation.

- **Where the gap closing comes from.** A decomposition shows that the effect works primarily through **higher GDR productivity growth** (rather than reduced FRG productivity growth), which fits a technology-transfer interpretation.
- **Aggregate counterfactual.** The paper concludes that industrial espionage had an overall *noticeable but quantitatively modest* effect on aggregate catch-up: the East-to-West German TFP ratio is about 21.8% in 1989, and would have been about 18.9% without espionage (about 13.3% lower relative to the realized ratio). Importantly, some high-tech sectors appear to have benefited much more than the aggregate number suggests.
- **Additional discipline.** Robustness checks (alternative keyword mappings, machine-learning sector assignment, alternative normalizations/weighting, sector trends) and placebo/randomization tests support the view that the main relationship is not a mechanical artifact of the panel structure.

7.2 Economic intuition

- **Espionage as a technology-diffusion shock.** Treat each “piece of information” as an increment to a sector’s usable knowledge stock. Then an increase in espionage inflows is

analogous to an outward shift in the sector’s effective technology set, which raises TFP growth in the recipient economy.

- **Absorptive capacity is the key complement.** The paper finds stronger effects in sectors where East Germany was *closer to the West German frontier* initially. The interpretation is that complex foreign knowledge is only productive if local engineers and firms can integrate it into existing processes and production networks; hence, a smaller initial technology gap implies higher absorptive capacity.
- **Quality matters more than quantity.** Evidence using the intelligence quality ratings suggests that productivity gains are driven disproportionately by a relatively small subset of high-value information items (a fat-tailed “impact distribution” of knowledge).
- **Espionage versus trade-based diffusion.** The paper tests whether espionage is more effective when East Germany faces higher import barriers (a larger import-intensity gap), but does not find strong differential effects across import-gap quartiles. A consistent interpretation is that espionage can substitute for missing trade-based diffusion, while at the same time effective implementation may require complementary access to imported inputs, leaving net effects similar across sectors.

7.3 Takeaway

Industrial espionage acted as a measurable technology-transfer channel that accelerated East German sectoral TFP catch-up, with effects strongest where absorptive capacity was high and where a few high-value intelligence items delivered outsized gains.